1) Codificador 4→2 (teclado → binário)

A1 = T2 + T3



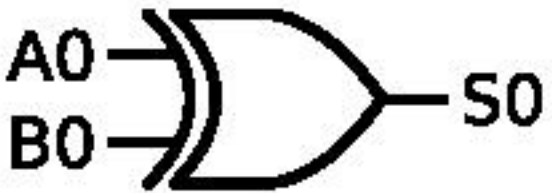
A0 = T1 + T3



T0 não precisa de porta (gera 00). Fazer 2 iguais: um p/ A, outro p/ B.

2a) Meio somador (bit 0)

S0 = A0 ⊕ B0

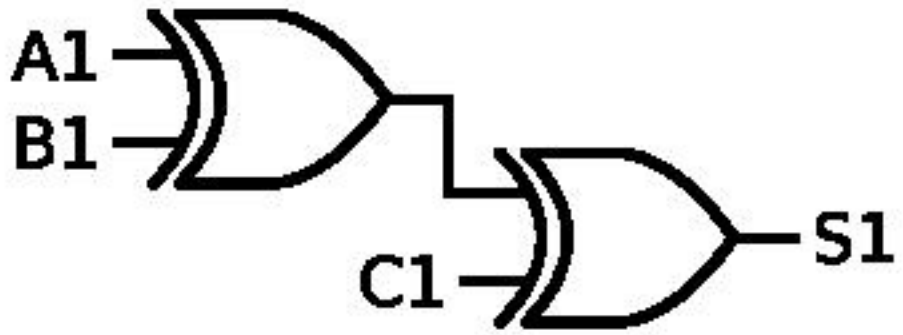


C1 = A0 · B0 (vai-um)

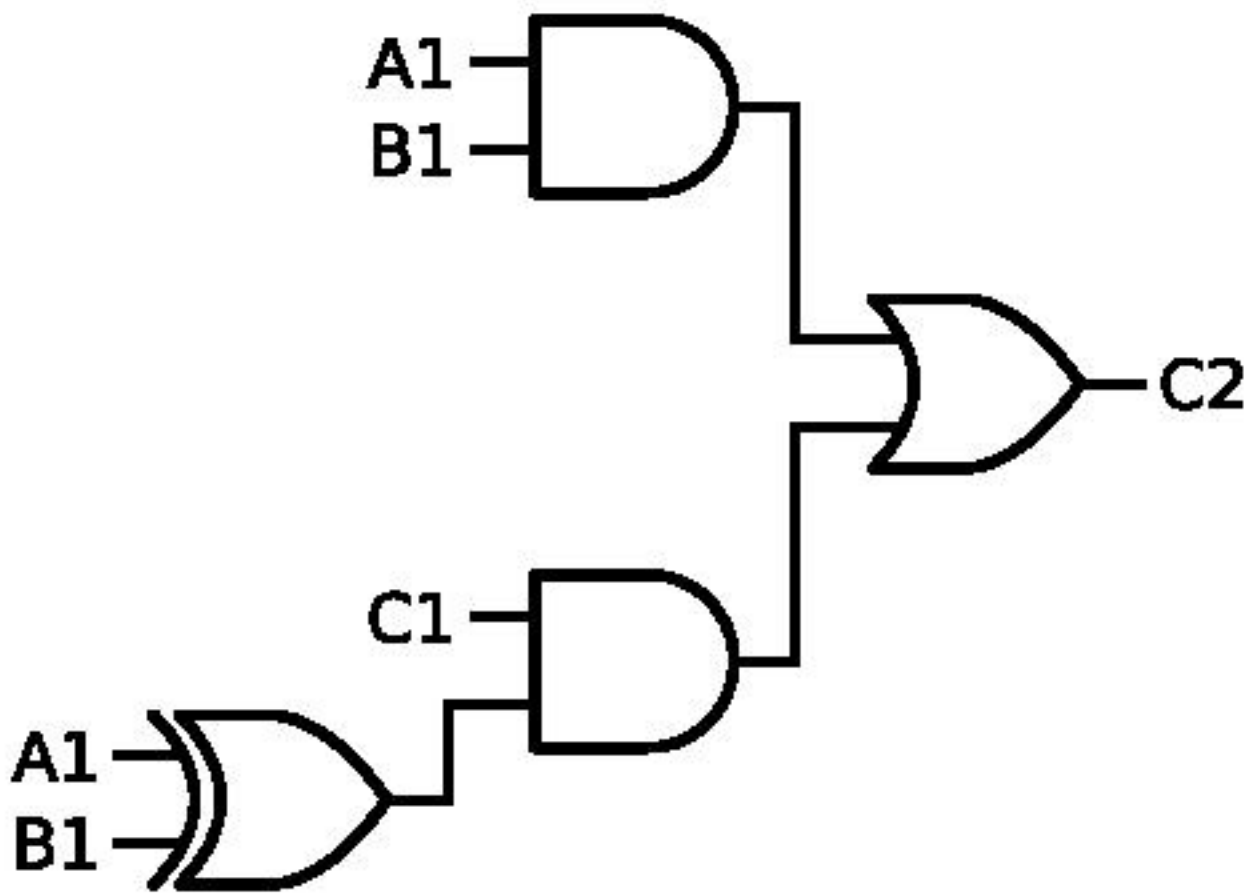


2b) Somador completo (bit 1)

S1 = A1 ⊕ B1 ⊕ C1



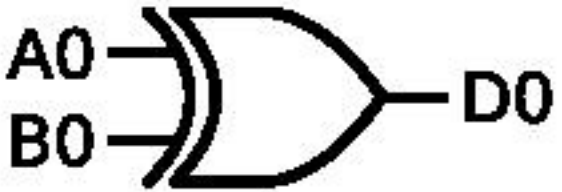
C2 = A1·B1 + C1·(A1 ⊕ B1)



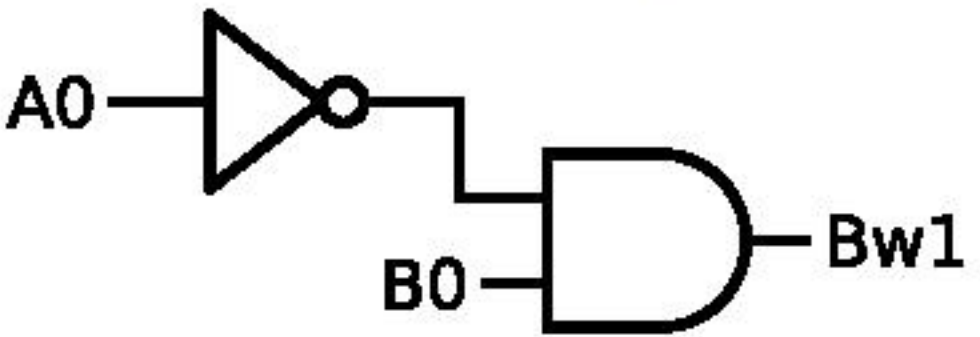
Nomes iguais = mesmo fio (repetido só pra organizar o desenho).

3a) Meio subtrator (bit 0)

D0 = A0 ⊕ B0

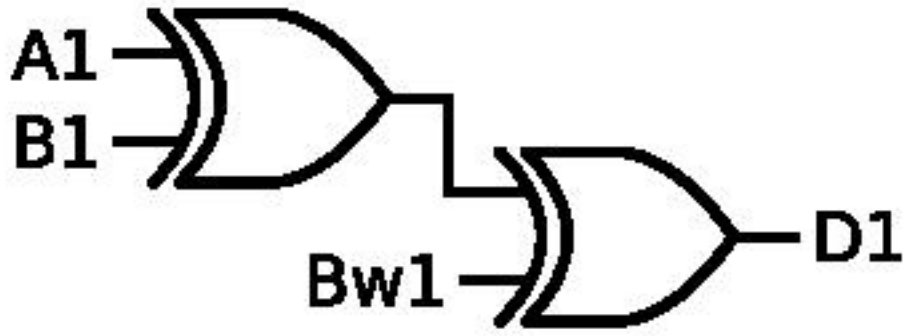


Bw1 = A0' · B0 (empresta 1)

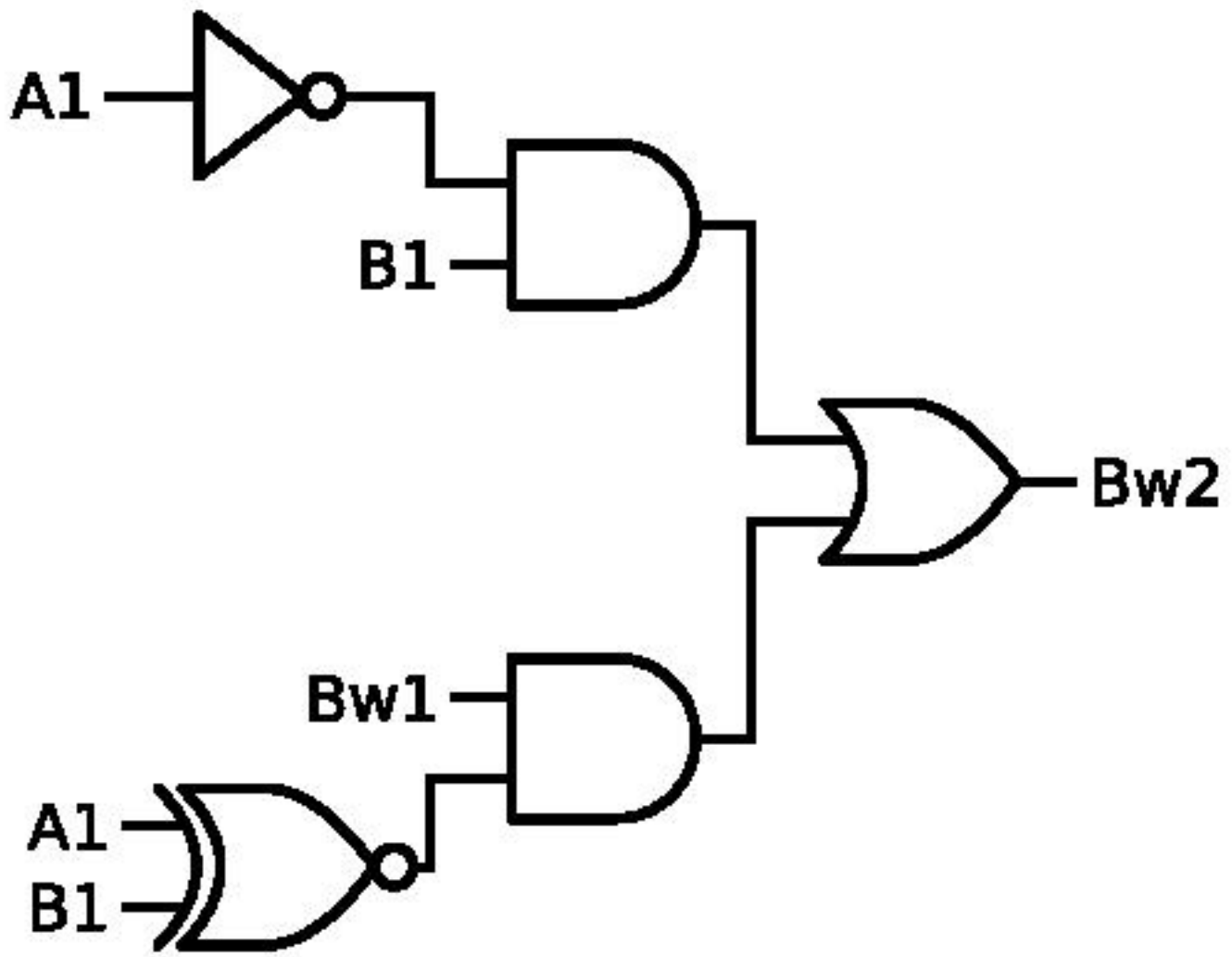


3b) Subtrator completo (bit 1)

D1 = A1 ⊕ B1 ⊕ Bw1



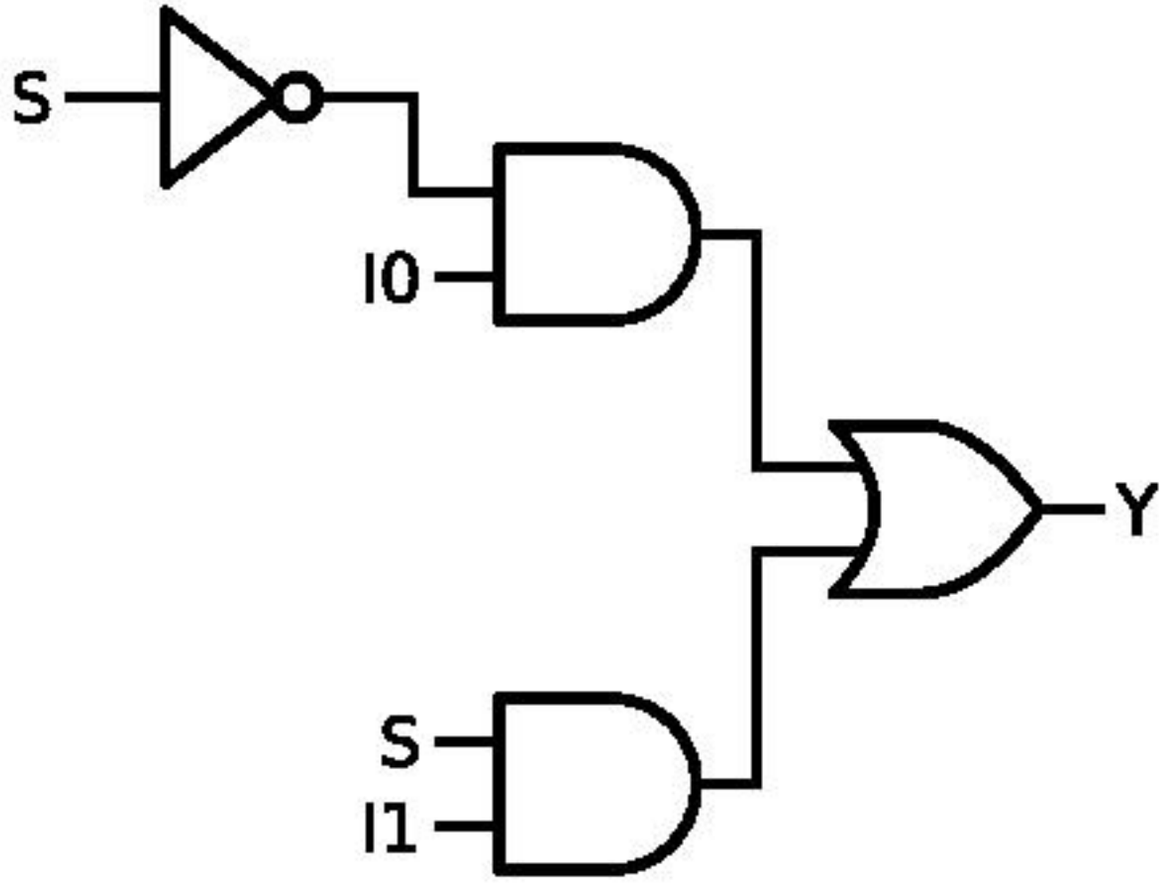
Bw2 = A1'·B1 + Bw1·(A1 ⊕ B1)'



Bw2 = 1 → A − B deu negativo (A < B). Nomes iguais = mesmo fio (repetido só pra organizar o desenho).

4) Multiplexador 2→1

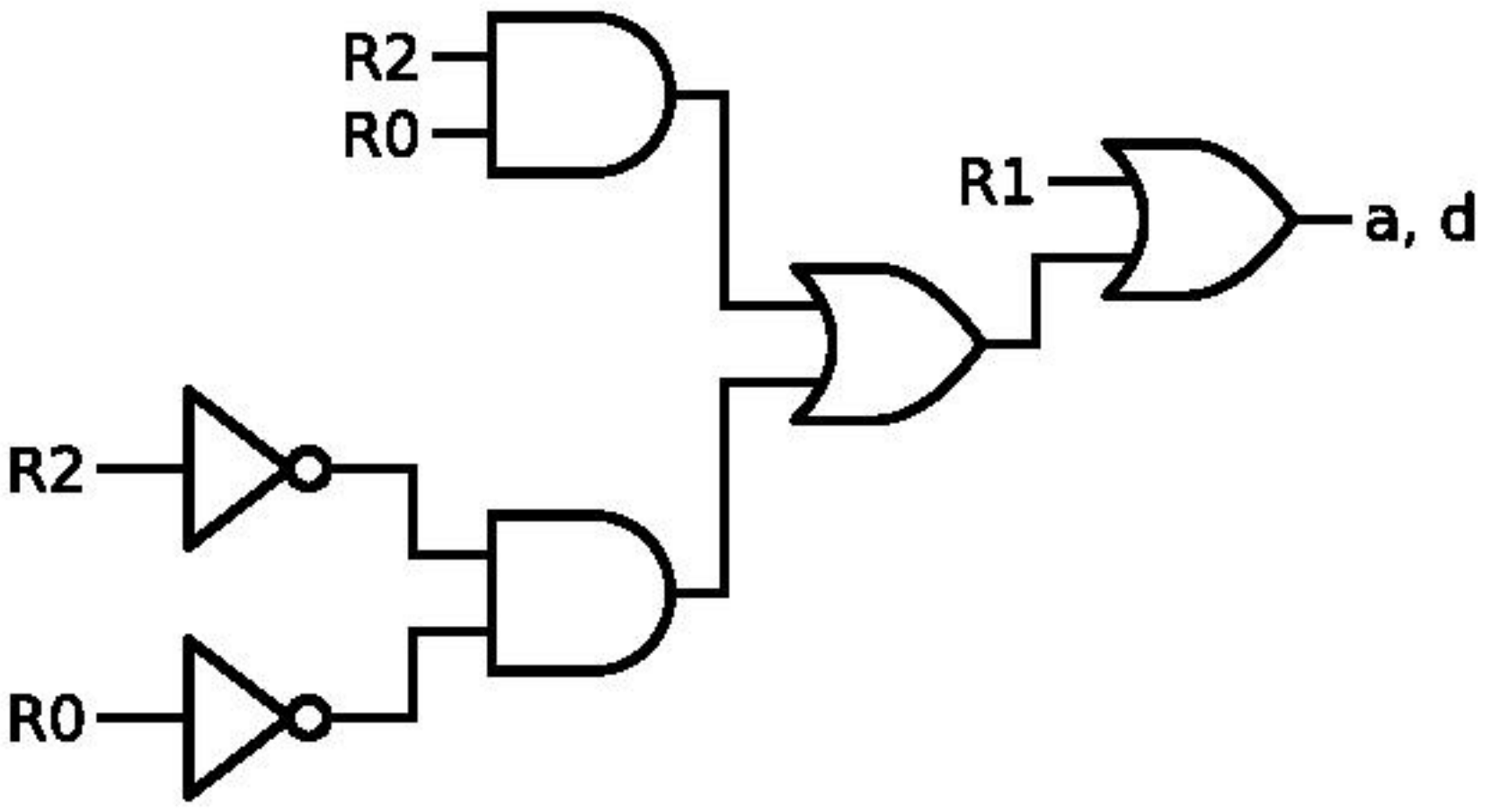
Y = S'·I0 + S·I1



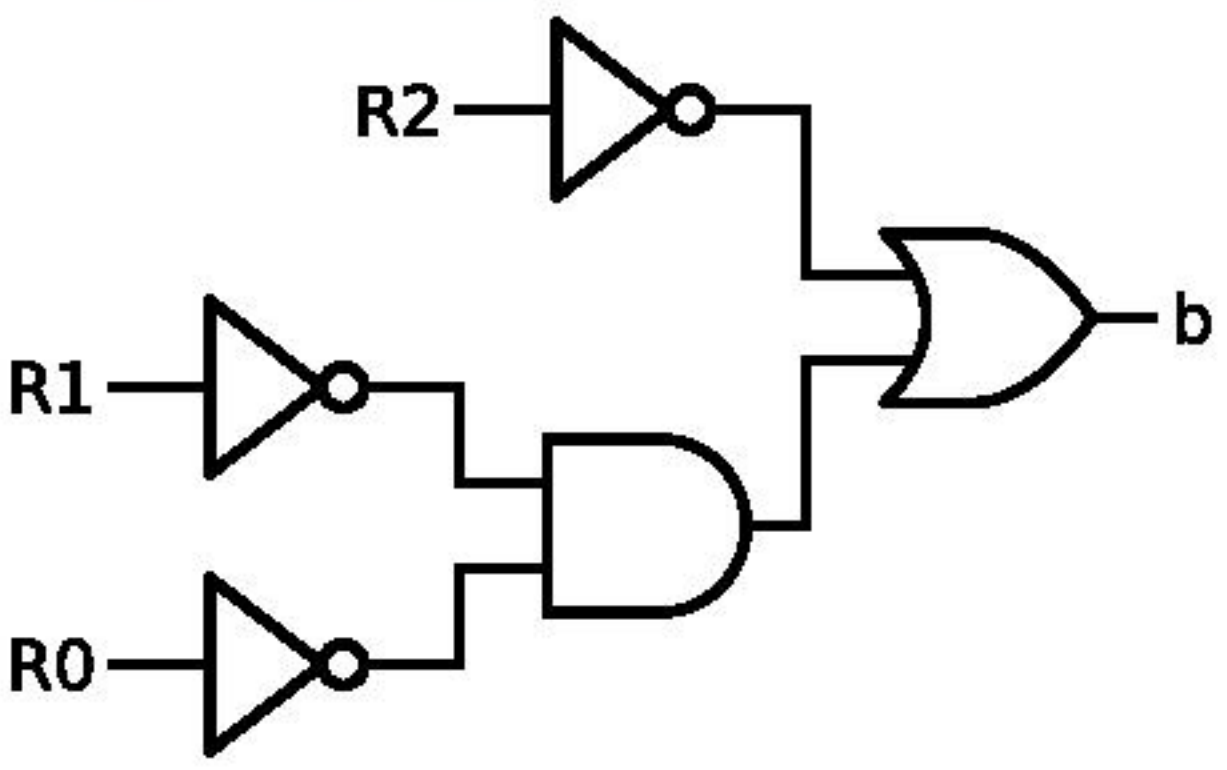
Usar 5 MUX: 2 com S = Bw2 (A−B ou B−A) e 3 com S = OP (soma ou subtração).

5) Decodificador binário → 7 segmentos

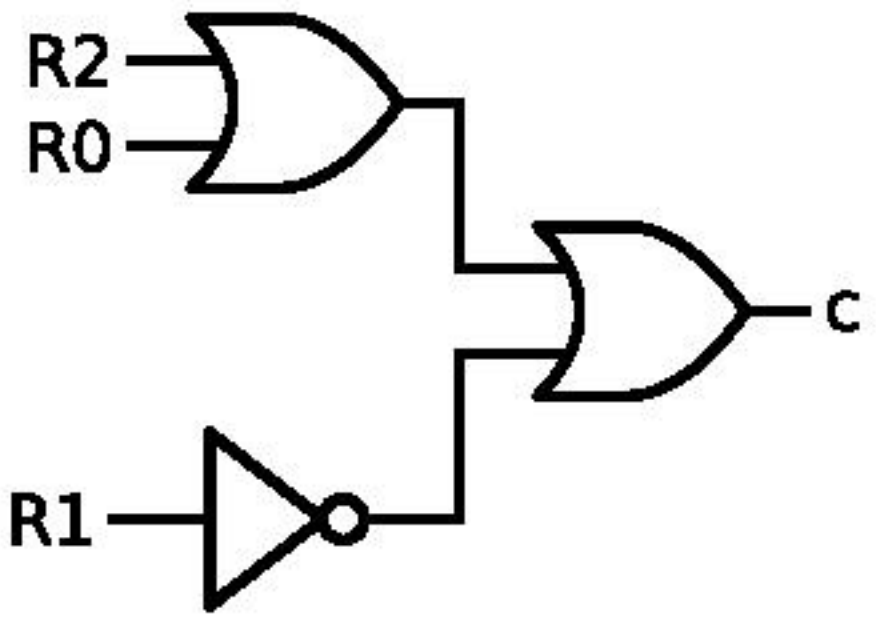
a = R1 + R2·R0 + R2'·R0' (d = a)



b = R2' + R1'·R0'

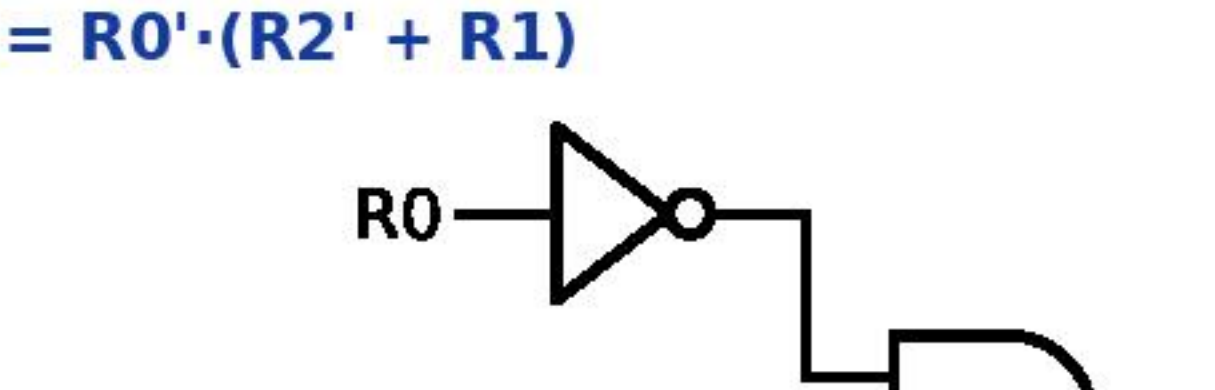


c = R2 + R1' + R0



e = R0'·(R2' + R1)

f = R2 + R1'·R0'



g = R2 + R1



O segmento d usa o mesmo fio do a.

6) Diagrama de blocos

